

VENKATRAMAN R

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venkatraman0400-blip.github.io/venkatraman-portfolio

Summary

Data Scientist & ML Engineer with a proven track record of architecting and deploying 8 production-grade machine learning and Generative AI systems. Expert across the full ML lifecycle, specializing in predictive modeling, secure LLM orchestration, and advanced statistical analysis. Advanced proficiency in building robust RAG pipelines, vector search indexing, and high-accuracy classifiers (AUC-ROC 0.97) featuring audit-compliant SHAP explainability. Adept at translating complex unstructured data into secure, scalable analytics applications.

Skills

Programming: Python, SQL

Math & Statistics: Statistics, Linear Algebra, Calculus, Probability, Algorithms, Hypothesis Testing

Machine Learning: Scikit-Learn, XGBoost, Random Forest, SMOTE, SHAP, Optuna, Cross-Validation, Feature Engineering

Deep Learning: TensorFlow, Keras, LSTM, Neural Networks, RNN, CNN

GenAI / LLMs: LangChain (LCEL), FAISS, RAG, OpenAI API, Groq Llama-3.3-70B, HuggingFace, Prompt Engineering, NLP

Analytics & Visualization: Pandas, NumPy, Power BI (DAX), Plotly, Matplotlib, Seaborn, Streamlit, SQLite, Parquet, Power Query

Tools: Git, GitHub, Jupyter, VS Code, Google Colab, Advanced Excel

Experience / Internship

Data Science Trainee

Jan 2026 – Jun 2026

Boston Institute of Analytics — Professional Certification Program

- Built FraudGuard, an XGBoost classifier with Optuna tuning (100 trials) and SMOTE balancing on imbalanced mobile-transaction data, reaching AUC-ROC 0.97, F1 0.92; encoded fraud-cost asymmetry via F2-score threshold optimization and added SHAP explanations for audit compliance
- Designed BankShield, an end-to-end Random Forest pipeline (EDA to feature engineering to SMOTE to threshold tuning to deployment) reaching AUC-ROC 0.97, precision 0.94, recall 0.89; identified transaction velocity as more predictive than amount and deployed a Streamlit app with single/batch CSV scoring

Projects

GenAI Dataset Q&A Application (AskMyData)

Core Tech Stack: Python, LangChain (LCEL), LLM Orchestration, Vector Databases (FAISS), SQL, Streamlit, Plotly

- Developed an autonomous LLM agent using LangChain LCEL that dynamically routes natural language queries to automated Pandas or SQL pipelines across multi-format datasets (CSV, Excel, JSON, SQLite).
- Implemented semantic context retrieval via FAISS vector indexing to map database metadata, boosting pipeline code-generation accuracy.
- Engineered a secure python execution sandbox featuring strict code sanitization, multi-turn conversational memory, automated error-handling loops, and native Plotly visualization generation.

SalesPulse — Power BI Sales Intelligence Dashboard

Power BI, DAX, Power Query, Star Schema

- Authored 44 custom DAX measures (YoY, MTD/QTD/YTD, customer LTV, profit margin) over a star schema on \$12.63M of Global Superstore data, with executive, regional, product, and customer-segment drill-throughs

SalesPulse Forecaster — LSTM 12-Month Sales Forecasting

TensorFlow, Keras, LSTM, SQLite, Streamlit

- Built a 2-layer deep learning LSTM network (128 to 64 units) generating recursive 12-month forecasts with confidence bands, backed by a SQLite pipeline and a Streamlit upload-to-export interface

DataHunt — Automated ETL Pipeline

Python, Pandas, Parquet, openpyxl

- Designed a modular pipeline (Extractor to Cleaner to Storage to Reporter) processing 51,000+ records and outputting Parquet and Excel simultaneously, with independently swappable modules

Vehicle Maintenance Dashboard — Fleet Analytics

Advanced Excel

- Built a fleet analytics dashboard over 50,000 vehicle maintenance records with KPI tracking and conditional-formatting-driven alerts

Education

Bachelor of Engineering (B.E.) — Automobile Engineering

2021 – 2024

Vels Institute of Science, Technology & Advanced Studies (VISTAS), CGPA: 7.25

Certifications

- Professional Certification in Data Science with Artificial Intelligence — Boston Institute of Analytics (Jun 2026)